

Crew Resource Management training

Module 1: planning and debriefing

Topics:

- Establishing a shared mental model (of what will happen)
- Planning before doing (including briefing): Stop, Assess, Plan.
 - ground, situation, mission, execution, administration, C3, risk assessment.
 - **Stop**: realize prep is needed and call the stop. Stop before you get into the action.
 - **Assess**: collect the necessary information from all crew members. Take some time for everyone to get oriented, if needed.
 - **Plan**: Decide what you'll do. Say it out loud and establish the shared mental model of what the next goal is. What should go wrong? When's next briefing?
 - **Ask: Any questions?**
 - **Decide: go/no-go decision.** Decide to accept the risk, or not.
- Situational awareness before starting and maintained during.
- Go/no-go decision
- Post-action review (more on this in module 4)

Exercise:

1. Do briefings:
 - a. ☐ When the scenario starts (once you get the needed info)
 - b. ☐ Before a new mission/task
 - c. ☐ Before a single tactical engagement
2. ☐ Do a post-action review after each individual action.
 - a. How did the action go? Ask each person.
 - b. Think about what to do better next time.
 - c. Think about one new thing you want to try next time.
3. ☐ Do a post-action review after the whole session.

Lessons for us office workers:

- Before doing things, ensure we have a good plan and shared mental model and explicit goal. Make sure everyone can contribute what they know. Write it down if it's big enough.
- Writing down "team strategy" is like doing a briefing, if it's actionable enough.
- Must make the plans/briefings concise enough so that people can follow. Too much text doesn't help everyone keep track.
 - We can't expect everyone to know everything, summarizing is an important skill.
 - I think good briefings and info is a skill and not easy to do.
- We often dive into ideas. We could more often have an explicit go/no-go decision (or several, as we iterate towards a solution). Sometimes doing nothing is better.
- We can do more retrospectives after we finish something.
- In summary, more explicit communication – shorter.

Module 2: Communication

Topics:

- Three-way communication and repeating back
 - (sender) **Say message** (like you normally would), (recipient) **Repeat message back in own words** (show you understand what it means), (sender) **Acknowledge correctness** (say "OK", "good", "correct", or similar, confirm it's correct – only now should recipient act).
- Standard procedures and phrases for consistent nomenclature.
- Verbalize what you are doing, even if it is according to plan
- Monitoring and call-outs (working/monitoring)
- Assertive statement 5-point template (see exercise point 2)
 - **Attention** (say the person's name), **concern** (generally what you are worried about), **root-problem** (what's causing the problem), **solution** (make a proposed solution), **buy-in** (say "ok, what do you think?" or similar).

Before playing:

1. Think about what situations often come up in the game. (battle, reloading, travel, etc.)
 - a. Think of the of the **standard operating procedures (SOPs)** for these situations.
 - b. Decide on **fixed names** for these.
 - c. **Write them down** (maybe multiple copies, and give to the stations that need them).
 - d. Engineer: program in presets for these (write them down).
2. [] Decide working/monitoring pairs for a task or station and work in these small teams. Maybe have two people assigned to one station.
 - a. Two people work together on a task or station. One is doing the action (small picture) and one is maintaining situational awareness and calling out important things they notice (big picture). Or the monitoring person handles communication with others.

Exercise:

1. [] [] Use three-way communication in at least two situations.
2. [] [] [] Use your SOPs at least three times.
3. [] Use the five-point assertive statement template to bring up an important situation.

Lessons for us office workers:

- Three-way communication. When doing critical work, repeat back the message to the sender.
- Even beyond that, it's not bad to repeat back the current task for confirmation.
 - Even when waiting for someone, you can say "I am currently waiting for XXX before I do YYY."
- When you see a problem that really needs to be addressed, you can use the 5-point assertive statement template to point it out.
- Often times our tasks are so small that having many people working on the same thing may not be worth it. In these cases, the one working/one monitoring strategy may be useful.

Module 3: Teamwork and decision-making

Topics:

- Input → Decision → Actions. How are decisions made.
- Timescale of decisions
- Maintaining situational awareness, OODA(Observe-Orient-Decide-Act) loop
- Checklists, task management, and clear task allocation
- “What if” scenarios and roleplay
- Workload management/prioritization

Before playing:

1. Think about the information flows. Adjust station physical locations into these “teamwork clusters”.
2. Improve your standard operating procedures (SOPs) from the previous module to include leadership and delegation of decision making.
 - a. For example, during X, the station X takes the lead in calling out what to do, Y and Z support them, and A and B maintain situational awareness.
 - b. Remember there can be multiple situations happening at the same time.
3. If you need, improve your checklists and SOPs:
 - a. Examples: what to check before an engagement. What to check before a mission. What to recover from after a mission. etc.
 - b. Decide who makes sure the checklists get used.

Exercise:

1. [] [] During two actions, the team delegates the leadership to a non-captain crew member.
2. [] Have at least one decision made by one person guiding a team decision (asking everyone’s info, deciding, doing it).
3. [] [] [] Use the SOPs or checklists you have made.

Lessons for us office workers:

- Input → Decision → Actions. We are good at doing things and good at finding stuff out. Sometimes we can be a bit slow on decisions, I think because we spread out responsibility too much. We need to make sure the decision-making is well-defined.
- Can we always even decide what decision needs to be made?
- Checklists are useful, if we can remember to make and use them. And know how to find them! Is this a lesson for our docs?
- There are several important roles we often don’t acknowledge/plan for during our own panics/work (we often think that “doing” is everything):
 - Monitoring person. Keeping an eye out for new information.
 - Output person. Either communicating with others, or doing things.
- Do we divide up and delegate tasks enough, or do everything by committee? Or by people working alone? Can we delegate information gathering, present in meeting to decide?

Module 4: Power distance and just cause reviews

Topics:

- Leadership styles (directive, supportive, participate, delegate)
- Authority gradient / power distance
- Speaking up when needed and asserting yourself
 - Report near misses to learn from them
- Retrospective analysis: not for blame, but for learning
 - **Facts:** Everyone has a chance to say any non-obvious facts or info that not everyone may know. Ensure everyone has a time to speak.
 - **Storyline:** Talk through together what happened in what order. Usually led by one person trying to put the story together, with help of course.
 - **Analysis:** Discuss why things happened as they did. What were the causes and effects? Root causes?
 - **Recommendations:** what to do different next time, or lessons that others should learn? Write them down somewhere people can find them.

Exercise:

1. Play again, putting all the previous lessons together it all together.
2. Take extra care to speak up when you notice something off, even if “you aren’t in change”.
3. [] After one battle, do an instant retrospective review

After playing, or after any actions that went wrong:

1. [] Do a retrospective review at the end of the session. Consider at least one thing that went wrong.

Lessons for us office workers:

- Are there more times when we should analyze what happened and learn for the future (and write it down...)?
- How can we have people speak up, without making a flood of information and getting stuck in discussions?
- What leadership styles are best for our team?
- Should we do better retrospectives to think about how things went?
- What problems do you see, that you haven’t yet spoken up about? (you don’t have to say now)

End of the day:

- Was this a good training and social day?
- What have we learned?
- Feedback?